Interrupt-Driven Temperature Monitoring System | STM32L476RG, Embedded C

- Designed and implemented an interrupt-driven temperature monitoring system on the STM32L476RG using Embedded C and a TMP36 sensor.
- Developed modular firmware for 12-bit ADC acquisition, EXTI interrupts, SysTick debouncing, and USART2/UART communication.
- Improved measurement reliability through 64-sample ADC averaging and an event-driven architecture optimized for low-power operation.
- Streamed real-time CSV-formatted temperature data to a PC via USART2 (115200 bps) and validated system timing and thermal response using MATLAB and a logic analyzer.